

Predicting players' performance in T20 cricket matches using Machine Learning approaches

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2019/SP/261

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A project report submitted in partial fulfillment of the requirements for the Course Unit
CSC406M6

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University of Jaffna

30 December 2024

Abstract

Player selection is a critical task in sports, particularly in cricket, where a team's performance hinges on various factors such as opposition, venue, and players' current form. The decision-making process involves meticulous analysis by the team management, coach, and captain to assemble the best playing XI from a squad of 15 to 20 players. Given the complexity of this task, machine learning has emerged as an effective tool for predicting player performance, thereby enhancing team selection and indirectly influencing match outcomes.

This study proposes a methodology for forecasting player performance in cricket using machine learning techniques, leveraging a dataset sourced from Kaggle. The proposed framework applies data preprocessing techniques, including mean filling for handling missing values and label encoding for converting categorical data into numerical format. Subsequently, Principal Component Analysis (PCA) is employed to identify the most influential features contributing to batting and bowling performance. The PCA results are further refined through correlation analysis to pinpoint features most relevant to the target variable.

To achieve accurate predictions, algorithms such as Linear Regression and Random Forest are utilized, extracting critical patterns from the dataset. By integrating these techniques, the study aims to build an automated and efficient system for player performance prediction, ultimately aiding team management in crafting an optimal playing XI. Additionally, a user-friendly GUI was implemented using Power BI to assist the team management in visualizing insights and making informed decisions. The results demonstrate the potential of machine learning in revolutionizing player selection strategies, fostering data-driven decisions, and elevating the overall performance of cricket teams.

Keywords: Cricket, Player Performance, Linear Regression, Random Forest

Acknowledgements

I am deeply grateful to Prof. A. Ramanan for his unwavering support and invaluable guidance throughout my research project on predicting players' performance in T20 cricket matches using machine learning approaches. Prof. Ramanan's expertise and dedication were instrumental in the success of this endeavor. His extensive knowledge in the field and his willingness to share his insights significantly enriched the quality of the research. Without his mentorship and encouragement, this project would not have been possible. Prof. Ramanan's commitment to my academic and professional growth has been a source of inspiration, and I am truly fortunate to have had the opportunity to work under his supervision.

Additionally, I would like to express my sincere appreciation to all the dedicated lecturers within the Department of Computer Science at the University of Jaffna. Their guidance, expertise, and academic support have immensely enriched this research endeavor.

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Acronyms

ML : Machine Learning

ODI : One Day International

T20I : Twenty-Twenty Internationals

GUI : Graphical User Interface

PCA : Principal Component Analysis

MSE : Mean Squared Error

RSE : R Squared Error

SVM : Support Vector Machine

RF : Random Forest

LR : Logistic Regression

DT : Decision Tree

KNN : K Nearest Neighbor

1. Introduction

1.1 Background

Cricket is one of the most popular sports globally, played and followed passionately by millions. Among its three major formats—Test matches, ODI, and T20—the shorter versions, ODI and T20, have garnered immense popularity among fans due to their fast-paced and thrilling nature. While Test cricket remains the oldest and most traditional form of the game, T20 cricket, which emerged in 2004, has revolutionized the sport by introducing a more dynamic and competitive format.

With the increasing competitiveness in cricket, administrators constantly seek innovative methods to enhance player performance. In the T20 format, where the margin for error is minimal, selecting the best playing XI becomes a critical and challenging task. The team management, coach, and captain meticulously analyze each player's characteristics and historical performance statistics to optimize team composition. A player's performance in cricket is generally assessed through their individual contributions, such as runs scored, strike rate, wickets taken, and bowling economy, among other metrics.

This research focuses on predicting player performance in T20 cricket using ML techniques. Specifically, it considers batting and bowling performances as the primary factors for making predictions. By employing ML algorithms, this study aims to provide an automated, data-driven approach to assist in selecting players for national teams. Additionally, the findings of this research hold significant potential value for the betting industry, where accurate performance predictions are highly sought after. Ultimately, this work seeks to contribute to the evolving landscape of cricket by demonstrating the applicability of machine learning in player performance forecasting.

Cricket research spans various areas, including game outcome prediction, player performance classification, and batting style or stroke classification. According to studies [4], team and performance analysis remains one of the least focused areas in cricket research. This highlights a significant gap in understanding and optimizing team selection and individual player contributions, especially in the T20 format. Addressing this gap, the main goal of this research is to focus on predicting players' batting and bowling performances using machine learning techniques. By doing so, this work aims to advance cricket analytics and provide data-driven solutions for improving team composition and performance.

Table 1 clearly illustrates the research areas in cricket, along with the percentage distribution of studies in each domain. From the table, it is evident that the performance of players is a less-explored area. This work aims to address that gap by focusing on player performance prediction, contributing to a more comprehensive understanding of this crucial aspect of cricket analytics.

Table 1: Research areas in cricket

Research Area	%studies
Game outcome Prediction	35
Players' performance classifications	16
Batting style/ Stroke classification	9
Bowling action/ Performance classification	9
Other	9
Umpire's decision/ gestures	6
Score prediction	5
Cricket Commentary/media	4
Pitch behaviour prediction	4
Team selection/Performance	3

1.2 Objective

This research focuses on selecting the most impactful features for player performance in batting and bowling. Additionally, it aims to develop a user-friendly GUI to assist team management in making efficient and data-driven team selection decisions, enhancing overall team performance in T20 cricket.

1.3 Outline

This research follows a structured framework to explore machine learning techniques for predicting player performance in T20 cricket. The Introduction establishes the significance of player performance prediction and the growing importance of data-driven approaches in cricket. The Background provides the theoretical foundation of cricket analytics, emphasizing batting and bowling metrics. The Objectives outline the study's focus on bridging research gaps and aiding player selection through ML models.

The Literature Review critically examines prior work on cricket performance prediction and related machine learning applications. The Methodology explains the dataset, preprocessing techniques like mean filling and label encoding, and the application of PCA and correlation for feature selection. It further details the implementation of algorithms such as Linear Regression and Random Forest.

A Comparison section evaluates model performance, highlighting their predictive accuracy and practical utility. The Conclusion summarizes findings and their implications, with Appendices offering supplementary information.

2. Literature Review

This section highlights key components drawn from related research papers in the field of cricket analytics. Researchers are increasingly utilizing machine learning and deep learning techniques to predict player performance and match outcomes. These studies analyze diverse datasets, including player statistics, match data, and team compositions, employing algorithms such as Random Forest, k-Nearest Neighbor, and shallow Convolutional Neural Networks. Critical components include effective feature selection methods, such as Principal Component Analysis and correlation analysis, as well as rigorous model validation through cross-validation techniques. The primary objective is to improve the accuracy of player performance predictions, aiding team selection and strategy development. These advancements have the potential to transform decision-making processes in cricket, providing data-driven insights for optimizing team performance and match results.

The study in [1] introduced a team composition-based approach to predict ODI cricket match outcomes using supervised learning algorithms. This approach focused on modeling team strength by analyzing the batting and bowling performances of individual players. Six attributes were used to train the bowler model, while five attributes were employed for the batsman model. The dataset utilized for training covered matches played between 2010 and 2014, restricted to the top nine ODI teams globally. Various machine learning algorithms were tested, and the k-nearest neighbor (k-NN) algorithm demonstrated the highest accuracy of 70.05%. This research highlighted the potential of data-driven models in understanding team composition and its impact on match outcomes, laying the foundation for future studies in cricket analytics. By leveraging individual player performance metrics, the study presented a compelling case for the application of machine learning in cricket strategy and decision-making processes.

The work in [2] presents a method to predict cricket match outcomes by modeling team performance based on individual player performances in previous matches. A shallow Convolutional Neural Network (CNN) with four layers was employed, as deeper CNN models typically require larger datasets. The dataset included 2,581 players and 474 ODI matches played between 2013 and 2017, with data extracted from the Cricinfo website. This study demonstrated the ability of machine learning models to analyze player contributions effectively and predict match outcomes with an accuracy of 70%. By focusing on a shallow CNN, the approach balanced model complexity with the limitations of available data, showcasing its practicality. The research highlights the value of leveraging player performance data to improve team strategy and decision-making, contributing to the growing field of cricket analytics and the use of machine learning in sports outcome predictions.

The study in [3] focuses on predicting the winners of ODI cricket matches using the Naïve Bayes (NB) algorithm. It utilizes a dataset with 15 features related to batting, bowling, team performance, and other relevant factors. The research aims to analyze the impact of these

variables on match outcomes. After applying the Naïve Bayes algorithm, the study achieves a prediction accuracy of 85.71%, demonstrating the effectiveness of the model. Data for the study was sourced from the ICC website, highlighting the value of historical match data in making accurate predictions for future ODI encounters.

The work described in [4] provides a comprehensive systematic review of published research over the past two decades on the application of machine learning (ML) in cricket. The review highlights that only 3% of studies focused on team selection, revealing a significant research gap in this critical area. To conduct the review, four major databases—Google Scholar, Science Direct, Scopus, and Web of Science—were utilized. A search using broader research domains was carried out in February 2022, employing the keywords “Applications of Machine Learning in Cricket.” All published work meeting these criteria was initially identified for analysis. This section aims to discuss the findings of the systematic review. First, it identifies the research areas in cricket where ML applications have been utilized, providing descriptive statistics about the reviewed publications. A detailed analysis follows in the research question section, where selected articles are reviewed to examine the types of feature extraction, ML techniques, and accuracy estimation methods employed. The next subsection focuses on the datasets and attributes used in these studies, highlighting their characteristics and scope. It also reviews data reduction and feature extraction techniques adopted in the analyzed research. The subsequent section presents an overview of frequently used ML algorithms, providing insights into their adoption and application in cricket. Finally, the review quantifies the accuracy of these ML techniques, offering valuable insights into their effectiveness. This systematic review underscores the potential of machine learning in cricket and the need for further exploration in team selection strategies.

The study in [5] focuses on predicting players' performance in ODI cricket matches, specifically analyzing runs scored by batters and wickets taken by bowlers. This work employs machine learning techniques, testing multiple algorithms to identify the most accurate model. For the analysis, the study considers seven attributes for batters and five attributes for bowlers, ensuring a comprehensive evaluation of performance metrics. Among the algorithms tested, Random Forest emerged as the most effective, achieving impressive prediction accuracies of 90.74% for runs scored and 92.25% for wickets taken. These results underscore the reliability and robustness of the Random Forest algorithm in handling complex datasets and extracting meaningful insights.

The study in [6] explores the application of machine learning algorithms in predicting cricket match outcomes. Using a dataset with 16 attributes, the work evaluates the performance of various algorithms, including ZeroR, Decision Tree, Random Forest, and XGBoost. Among these, XGBoost demonstrated the highest accuracy, achieving 85.48%. The data for this analysis was sourced from the Cricsheet website, which provides detailed cricket match information. This study highlights the effectiveness of advanced ML algorithms like XGBoost in analyzing complex cricket datasets and accurately predicting match outcomes, emphasizing their potential for enhancing strategy and decision-making in cricket analytics.

The study in [7] implements a data mining system to predict the winning team in ODI cricket matches. It focuses exclusively on matches played between ICC full member teams,

encompassing 12 teams from 2015 to 2020. The research utilizes Logistic Regression and Multi-Layer Perceptron (MLP) algorithms to build predictive models. Among the techniques evaluated, both Logistic Regression and MLP achieved the highest accuracy of 70%. The dataset, containing detailed match information, was extracted from the Cricinfo website. This study demonstrates the potential of data mining techniques in predicting match outcomes and contributing to decision-making in cricket strategy.

Author(s) & Year	Task	Methodology	Dataset	Performance
1. Madan Gopal and Vikram Pudi (2016) [1]	Predicting the outcome of ODI cricket matches using a supervised learning approach from a team composition perspective.	Support Vector Machine, Random Forest, Logistic regression, Decision Tree, K-NN.	ESPN cricinfo site. Consider five and six attributes to create batsmen and bowler model respectively.	Accuracy SVM ->58.56% , RF -> 63% , LR ->64% , DT -> 66%, KNN -> 70.05%.
2. Manivannan and Kausik (2019) [2]	Predicting the outcome of cricket matches by modelling the team performance based on the performance of its players in other matches.	Feature Encoding, Shallow CNN (use of 4 layers because of limited number of data in training dataset)	ESPN cricinfo site Dataset includes 2581 players and 474 ODI matches from 2013 to 2017.	Accuracy Outcome of a cricket match can be predicted with accuracy 70%.
3.I.Wickramasinghe [2020] [3]	Predict the winner of ODI cricket game. [12 Teams]	Used Naïve Bayes approach with confusion matrix and K-cross validation.	ICC website From 2018 to 2020 Dataset includes 4 categories Batting, Bowling, Team and Other. This works includes 14 attributes.	Accuracy 80.71%

4. I.Wickramasinghe [2022] [4]	A systematic review on applications of machine learning in cricket.	By identifying all published work using 4 databases, selected most relevant publications while removing the duplicates.	From 2001 to 2021. 59 publications. Google Scholar, Science direct, Scopus, Web of Science.	
5. Kalpdurum Passir & Niravakumar Pandey [2018] [5]	Predicting players' performance in ODI matches	Used Naïve Bayes, SVM, Random forest and decision tree	Cricinfo site Considered 7 attributes for batters and 5 attributes for bowlers.	Random forest shows the best accuracy in terms of runs and wickets 90.74% and 92.25%
6. Fahim et al [2020] [6]	Predicting the outcome of international cricket matches.	ZeroR, Decision tree, Random Forest, XGBoost applied to dataset.	Cricsheet site From 2004 to 2018 Considered 16 attributes.	XGBoost algorithm perform 85.48% as accuracy.
7. Dinithi et al [2021] [7]	Analyze and built the best outcome prediction model for ODI matches. (Implement a data mining System)	Logistic Regression, MLP, Gaussian Naïve bayes, Linear Discriminant Analysis used as ML algorithms to predict the accuracy of train and test data.	Cricinfo site From 2015 to 2020	Logistic regression and MLP measure the accuracy as 70%.

Table 2: Summary of the Literature review

3. Background

3.1 Random Forest

Random Forest Regression is an ensemble learning method that combines the predictions of multiple decision trees to improve the overall accuracy and robustness of the model. In the case of regression, Random Forest Regression predicts a continuous outcome.

Here's an introduction to Random Forest Regression with a simple figure:

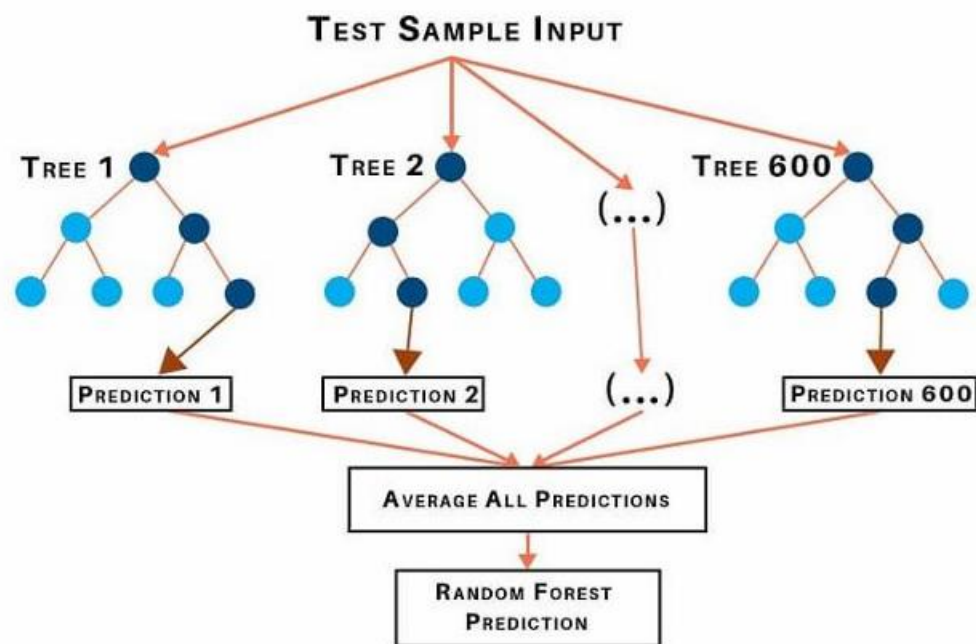


Figure 1: Random forest model example

Component of the figure:

1. Data points:

- The dots represent the data points with their corresponding features and target values. In regression, the target values are continuous.

2. Decision Trees:

- Each tree is constructed using a random subset of the features and a random subset of the training data. This randomness helps reduce overfitting.

3. Random subsampling:

- At each node of the tree, a random subset of features is considered for making a split. This ensures that the trees are diverse and capture different aspects of the data.
4. Ensemble of trees:

- The Random Forest consists of multiple decision trees. Each tree makes its prediction, and the final prediction is obtained by averaging (for regression) the predictions of all the trees.

5. Aggregated prediction:

- The final prediction is the average (or another aggregation method) of the prediction from all the trees. This ensemble approach helps improve accuracy and generalization.

How Random Forest Regression Works:

1. Bootstrapped sampling:

- Random subsets of the training data are created through bootstrapped sampling (sampling with replacement). Each tree is trained on one of these subsets.

2. Random feature selection:

- At each split in the decision tree, a random subset of features is considered for making the split. This introduces diversity among the trees.

3. Tree construction:

- Each tree is grown until a predefined stopping criterion is met, such as reaching a certain depth or having a minimum number of samples in a leaf node.

4. Prediction aggregation:

- For regression, the predictions of all the trees are averaged to obtain the final prediction of the Random Forest.

In summary, Random Forest Regression is a powerful ensemble learning technique that leverages the strength of multiple decision trees to make accurate predictions in regression tasks. The diversity among the trees helps mitigate overfitting and enhances the model's generalization capabilities.

3.2 Linear Regression

Linear Regression is a simple yet powerful statistical method used for predicting continuous outcomes based on one or more input features. It assumes a linear relationship between the input features (independent variables) and the target variable (dependent variable). Linear Regression aims to find the best-fit line (or hyperplane in higher dimensions) that minimizes the error between the predicted and actual target values.

Here's an introduction to Linear Regression with a simple figure:

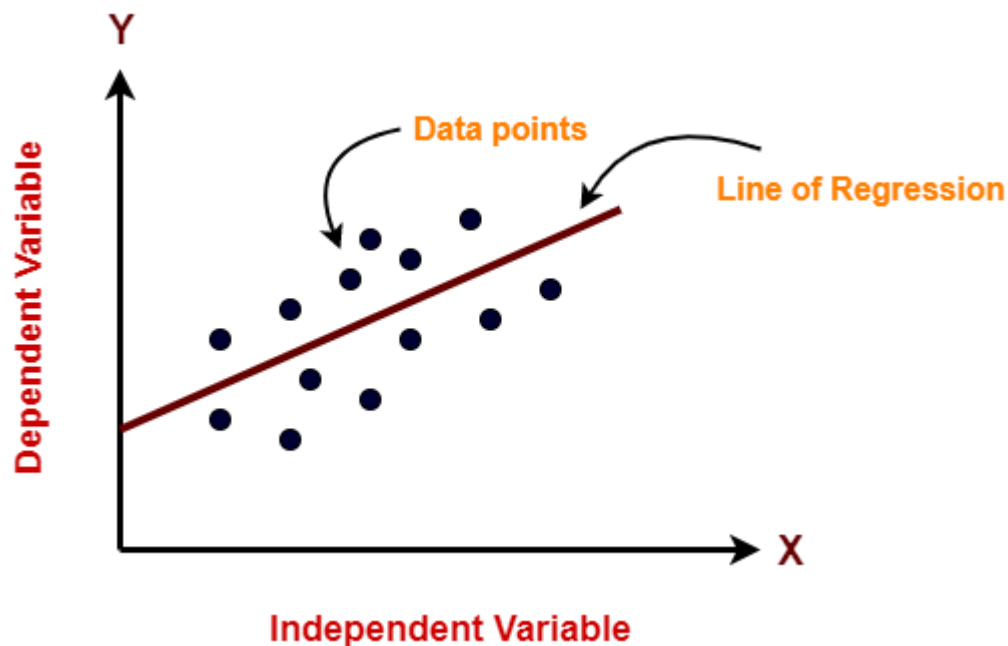


Figure 2 : Linear Regression Model Example

Components of Linear Regression:

1. Data Points:

- The dots represent the observations in the dataset, with their corresponding features (independent variables) and continuous target values (dependent variable).

2. Model Equation:

- The model follows the equation:
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon$$

Here, Y is the predicted target variable, β_0 is the intercept, $\beta_1, \beta_2, \dots, \beta_n$ are the coefficients for the features $X_1, X_2, \dots$, and ϵ error term.

3. Error Metric:

- The error is typically measured using Mean Squared Error (MSE) to quantify how far predictions are from actual values.

4. Optimization (Finding Best Fit):

- Linear Regression uses the method of Ordinary Least Squares (OLS) to minimize the sum of squared differences between actual values and predicted values, ensuring the best-fit line is as close as possible to the data points.

How Linear Regression Works:

1. Data Preparation:

- The input data is prepared by ensuring no missing values, scaling features (if necessary), and checking for multicollinearity among features.

2. Model Training:

- The model identifies the coefficients $\beta_0, \beta_1, \dots, \beta_n$ by minimizing the sum of squared residuals:

$$\text{Residual} = (Y_{\text{actual}} - Y_{\text{predicted}})^2$$

3. Making Predictions:

- Using the learned coefficients, the model predicts continuous target values based on the input features using the linear equation.

4. Assumptions:

- Linear Regression makes several assumptions:
 - **Linearity:** The relationship between the input features and the target variable is linear.
 - **Independence:** Observations are independent of each other.
 - **Homoscedasticity:** The variance of errors is constant across all levels of the independent variables.
 - **Normality:** Errors are normally distributed.

5. Evaluation:

- The model's performance is evaluated using metrics like R-squared, and MSE which indicate how well the predictions match the actual values.

3.3 Regression Evaluation Metrics

Regression evaluation metrics are crucial for assessing the accuracy of models predicting continuous outcomes. Mean Squared Error (MSE) highlights larger errors by squaring differences, while Root Squared Error (RSE) offers a scaled version of MSE for easier interpretation. These metrics are essential for evaluating, comparing, and refining regression models to ensure optimal predictive performance.

3.3.1 Mean Squared Error

Mean Squared Error (MSE) is a regression metric that calculates the average of the squared differences between predicted ($predicted_i$) and actual ($actual_i$) values.

The formula for MSE is given:

$$MSE = \frac{1}{n} \sum_{i=1}^n (actual_i - predicted_i)^2$$

MSE provides a measure of prediction accuracy that emphasizes larger errors due to the squaring operation. It is commonly used in regression analysis to assess the overall fit of a model by considering both small and large deviations from the actual values.

3.3.2 R-Squared Error

R-squared (coefficient of determination) is a regression metric that quantifies the proportion of variance in the dependent variable ($actual_i$) explained by the independent variables in the model.

The formula for R-squared is:

$$R^2 = 1 - \frac{\sum_{i=1}^n (actual_i - predicted_i)^2}{\sum_{i=1}^n (actual_i - \overline{actual})^2}$$

where $\overline{actual}$ is the mean of the actual values. R-squared values range from 0 to 1, where 1 indicates a perfect fit and lower values indicate less explanatory power. It is a valuable metric for assessing how well the model captures the variability in the target variable and is commonly used in regression analysis for model evaluation.

4. Methodology

4.1 Dataset

The data set is found from Kaggle (<https://www.kaggle.com/datasets/bhuvaneshprasad/all-t20-internationals-dataset-2005-to-2023>) which were obtained from cricinfo site. It contains six types of datasets such as Player's data, Batting data, Bowling data, Wickets data, Matches data, Partnership data. In this work mostly focus on Batting and bowling figures which includes 51738 and 28138 data respectively. This dataset includes data spanning from 2005 to 2023.

Features of the Dataset

Batting	Bowling
Match ID	Match ID
Innings	Innings
Teams	Teams
Batsman	Opposition
Runs	Bowler
Balls	Overs
Fours	Balls
Sixes	Maidens
Strike Rate	Conceded
Is Out	Wickets
Wicket Type	Economy
Fielders	Dots
Bowler	Fours
	Sixers
	Wides
	No Balls

Table 3 : Features of the Dataset

4.2 Methodology

Before evaluating the models for both batting and bowling performance, the following preprocessing and feature selection steps were applied to each section of the dataset.

Step 1: Pre-processing Step - In this initial phase, we addressed missing data by applying mean filling, where missing values in the dataset were replaced with the mean of each respective column. This ensured that the dataset remained complete and consistent for further analysis. Additionally, a label encoder was used to convert categorical data into numerical data, with each category assigned a unique integer. This transformation was necessary for machine learning algorithms, which require numerical input for effective processing.

Step 2: Feature Selection Step - Following the pre-processing, Principal Component Analysis (PCA) was applied to the entire dataset to reduce dimensionality and select the most influential features. PCA helped identify the key features contributing the most to the variation in the data, particularly focusing on the components that affected batting and bowling performance. The primary feature extracted from PCA, referred to as PCA1, was selected for further analysis. Once the important feature was identified, correlation analysis was performed to assess the relationships between PCA1 and other features in the dataset. The features that were most highly correlated with PCA1 were selected for predicting the target variable. This approach ensured that only the most relevant and influential features were used to make predictions, enhancing the accuracy and efficiency of the machine learning models used for both batting and bowling performance evaluation

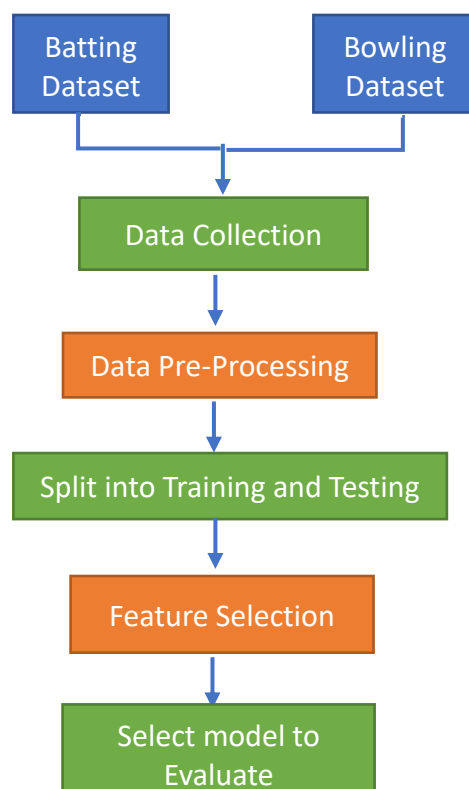


Figure 3: Methodology

The model evaluation process uses two methodologies. both method are under regression analysis.

They are;

1. Random Forest Regression
2. Linear Regression

4.2.1 Batting Section

For predicting batting performance, the methodology begins with data preprocessing to prepare the dataset for analysis. Missing values were handled using mean imputation, and categorical variables were transformed into numerical format using a Label Encoder. After preprocessing, Principal Component Analysis (PCA) was applied to identify the most influential feature affecting batting performance. The PCA revealed that "Runs" is the most critical feature. Subsequently, correlation analysis was performed to identify the features most strongly correlated with "Runs." The analysis showed that "Balls", "Fours," "Sixers," and "Strike Rate" are the features most strongly related to "Runs." These correlated features were then utilized as predictors in the machine learning models, enabling the prediction of batting performance. This approach ensures that the focus is on the features most relevant to a player's batting success, thereby improving the accuracy and reliability of the predictions.

```
pca1_eigenvector = eig_vecs[:, 0]
feature_names = X.columns
max_index = np.argmax(np.abs(pca1_eigenvector))
most_influential_feature = feature_names[max_index]
pca1_contribution_value = np.abs(pca1_eigenvector[max_index])
print(f"The most influential feature contributing to PCA 1 is: '{most_influential_feature}' with a contribution of {pca1_contribution_value:.4f}")
```

The most influential feature contributing to PCA 1 is: 'runs' with a contribution of 0.5148

Figure 4 : Selecting most influential feature for batsman

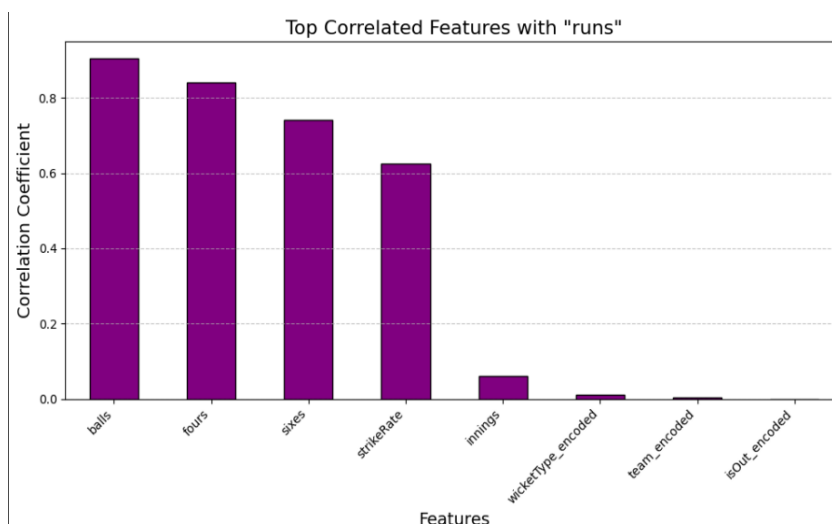


Figure 5:
Correlation
with runs

4.2.2 Bowling Section

For bowling performance, the methodology similarly begins with data preprocessing, where missing values were filled using mean imputation, and categorical data were converted into numerical form using a Label Encoder. Following preprocessing, PCA was applied to determine the most influential feature for bowling performance. The PCA identified "Conceded" (runs conceded by a bowler) as the key feature.

Next, correlation analysis was conducted to identify the features most closely associated with "Conceded." The analysis revealed that "Fours", "Sixers," "Balls," and "Economy" are the features most correlated with "Conceded." These correlated features were subsequently used as predictors in the machine learning models to predict bowling performance. This methodology ensures that the models focus on the critical aspects of a bowler's performance, enabling accurate and insightful predictions for team selection and strategy formulation.

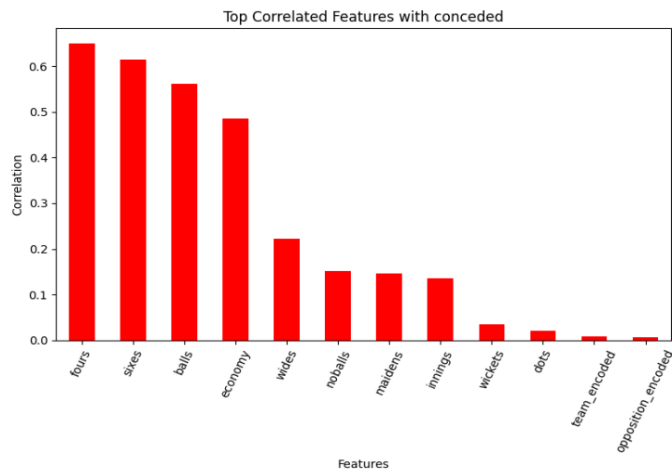


Figure 6 : Correlation with conceded

4.2.3 GUI Implementation

Power BI is used for the visualization.



Figure 7 : GUI Implementation

5. Results

The performance of Linear Regression and Random Forest was evaluated for both batting and bowling sections using RSE and Mean Squared Error MSE metrics on the training and testing datasets. For batting, Linear Regression achieved an RSE of 0.9752 and an MSE of 6.2218 on the training dataset, and an RSE of 0.9551 with an MSE of 6.6149 on the testing dataset. In contrast, Random Forest performed better, with an RSE of 0.9873 and an MSE of 0.9965 on the training dataset, and an RSE of 0.9941 and an MSE of 1.5066 on the testing dataset, demonstrating higher accuracy and lower error. For bowling, Linear Regression showed an RSE of 0.9015 and an MSE of 10.9456 on the training dataset, while the testing dataset results were RSE of 0.8898 and MSE of 9.8721. Random Forest outperformed Linear Regression in bowling as well, achieving an RSE of 0.9456 and an MSE of 0.0054 on the training dataset, and an RSE of 0.9098 with an MSE of 5.1478 on the testing dataset. These results confirm that Random Forest is the more accurate and robust model for predicting player performance in both batting and bowling.

Section	Algorithms	Training Dataset		Testing Dataset	
		RSE	MSE	RSE	MSE
Batting	Linear Regression	0.9752	6.2218	0.9551	6.6149
	Random Forest	0.9873	0.9965	0.9941	1.5066
Bowling	Linear Regression	0.9015	10.9456	0.8898	9.8721
	Random Forest	0.9456	0.0054	0.9098	5.1478

Table 4: Results

6. Comparison

In comparing the two models, Linear Regression and Random Forest—across the evaluation metrics used in this study, it is evident that Random Forest emerges as the most robust performer. Random Forest consistently demonstrates superior predictive accuracy with the lowest Mean Squared Error and R-Squared Error among the models. These results indicate that Random Forest is more effective at minimizing prediction errors and capturing the complex relationships within the dataset.

In contrast, Linear Regression, while providing reasonable performance, lags behind Random Forest in terms of both metrics. This can be attributed to its assumption of linearity, which may not fully capture the intricate patterns in the data. Overall, based on the evaluation metrics of MSE and RSE, Random Forest stands out as the most effective model for predicting player performance in T20 cricket matches. Its versatility and ability to handle non-linear relationships make it a superior choice for this specific task.

7. Conclusion

In this study, we focused on predicting player performance in T20 cricket matches based on players' batting and bowling statistics. However, other factors that can significantly influence performance, such as weather conditions or the nature of the wicket, were not included in the analysis due to data unavailability. These factors are often crucial in real-world scenarios, and their inclusion could enhance the accuracy of predictions.

The accurate selection of players for each match is a vital part of a team's strategy. Predicting key performance indicators, such as the number of runs a batsman might score or the number of runs a bowler may concede, can guide team management in making informed decisions.

In our study, we used Random Forest as the primary model for predicting both batting and bowling performance. Among the models tested, Random Forest emerged as the most accurate, achieving 98% accuracy for predicting batting performance and 90% for bowling performance. These results align with findings from previous work [5], where Random Forest also demonstrated high accuracy, with 90.74% in predicting runs and 92.25% in predicting wickets. These outcomes highlight the model's strong ability to handle complex relationships in the data and make precise predictions, showcasing its suitability for player performance prediction in cricket.

8. Appendices

Here is the Google Drive link related to my research.

https://drive.google.com/drive/folders/18uJWklHNnvHFAcXN9y2oSyyFqaQpGVX8?usp=drive_link

9. References

- [1]. Jhanwar, M. G., & Pudi, V. (2016). Predicting the Outcome of ODI Cricket Matches: A Team Composition Based Approach. *MLSA@ PKDD/ECML*, 78.
- [2]. Manivannan, S., & Kausik, M. (2019). Convolutional neural network and feature encoding for predicting the outcome of cricket matches. In *2019 14th Conference on Industrial and Information Systems (ICIIS)* (pp. 344-349). IEEE.
- [3]. Wickramasinghe, I. (2020). Naive Bayes approach to predict the winner of an ODI cricket game. *Journal of Sports Analytics*, 6(2), (pp. 75-84).
- [4]. Wickramasinghe, I. (2022). Applications of machine learning in cricket: a systematic review. *Machine Learning with Applications*, 10, 100435.
- [5]. Passi, K., & Pandey, N. (2017). Predicting players' performance in one day international cricket matches using machine learning. *Computer Science & Information Technology (CS & IT)*.
- [6]. Shakil, F. A., Abdullah, A. H., Momen, S., & Mohammed, N. (2020). Predicting the result of a cricket match by applying data mining techniques. In *Software Engineering Perspectives in Intelligent Systems: Proceedings of 4th Computational Methods in Systems and Software 2020, Vol. 2 4* (pp. 758-770). Springer International Publishing.
- [7]. Hasanika, D., Dilhara, R., Liyanage, D., Bandaranayake, A., & Deegalla, S. (2021). Data mining system for predicting a winning cricket team. In *2021 IEEE 16th International Conference on Industrial and Information Systems (ICIIS)* (pp. 92-97). IEEE.